

STATISTICAL EQUIVALENCE TESTING AS A TOOL TO ANSWER THE REAL QUESTION IN STABILITY TESTING FOR EQA SAMPLES

Lobna Bouacida¹ • Mohamed Rida Soumali¹ • Wim Coucke¹

1. Quality of laboratories, Sciensano, Brussels, Belgium

- The current approach mentioned in ISO 13528 is prone to wrong conclusions. A simple t-test with a higher limit of samples is not an option neither because this test does not respond to the question of proving that the difference between two measurements is sufficiently low.
- Equivalence testing is an elegant solution, but needs a limit that defines when the means of the two measurement times are close enough to each other. For this reason, an algorithm was developed to draw the link between the maximum difference between the two means and an allowable, very slight, increase of the risk of wrongly evaluating EQA results.
- This approach can be applied to every scheme for quantitative results, and, if laboratories report the time of analysis, can even be used based on the reported EQA results and hence, does not need any extra laboratory analysis.

Stability testing for EQA samples is important to proof that the analyte under investigation has the same assigned value during the analysis period. Therefore it is required by ISO 17043, that refers for details to ISO 13528. The algorithm proposed by ISO 13528 is vulnerable to wrong conclusions, though.

It can be easily proven that, if the minimum required samples for stability testing are taken, namely two samples, the probability to accept the samples for stability is very low, in the range of 45%, even when the parameter value does not change during the analysis period.

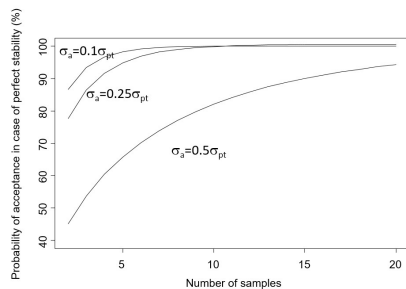


Fig 1. In the ISO 13528 method, the probability of acceptance in case of perfect stability remains low, even after increasing the number of samples tested.

An alternative simple test for stability testing is proposed. It is based on equivalence testing and minimising the number of wrong EQA evaluations.

MATERIAL

- Results from the **Belgian external quality assessment scheme for haematology**.
- Investigated parameters: red blood cells (RBC), white blood cells (WBC), haemoglobin (HB), haematocrit (HCT), mean corpuscular volume (MCV), thrombocytes and reticulocytes.
- Only the results analysed on a Sysmex system are reported here.
- The laboratories are instructed to analyse the samples within 2 days from the date of shipment (day 0). They must also report the date of the analysis.

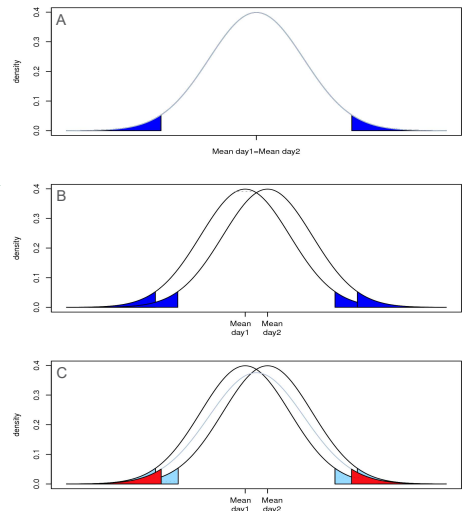
METHOD

- Stability testing requires the proof that the difference in mean analyte concentration between two distinct time points is lower than a certain limit.
1. **Calculation of the maximum allowed difference between the means of the two time points**
 - by assessing the **change in probability of falsely flagging** a result from a well performing laboratory if the results are analysed without distinction between analysis date (fig 2).
 - This probability is set to arbitrary maximum values of **5%points for Q-scores** and **0.2%points for Z-scores**.
 - False flagging means that a well performing laboratory is flagged for its Z-scores or for its fixed deviation with respect to the assigned value.

Fig 2. A: There is 100% stability. The means of the results obtained on the two days coincide. The chance of falsely flagging well performing laboratories (blue area) is the same for the two days, whether the data would be analysed differently or together.

B: Due to instability, the mean of the data obtained on the second day has shifted to the right. The limits beyond which well performing laboratories are falsely flagged are different if the results would be evaluated according to day of analysis.

C: The same case as the graph B, but now the effect of joining the data obtained on the two days is shown. The red area indicates the limits beyond which well performing laboratories are flagged if no distinction between analysis date is made. The light blue zones show the probability of changed flagging status: some results have now a higher chance of being flagged, some results have a lower chance of being flagged. The larger the difference between the two means, the larger these areas. Stability is now defined as the case where those blue surfaces remain minimal.



2. Equivalence testing

- performed by comparing the confidence interval around the difference between the two means with the limit of the maximum allowed difference (fig 3).

RESULTS

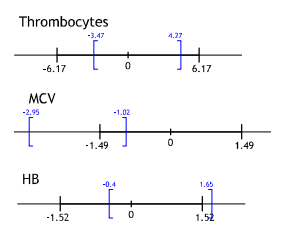
Tab 1. Application of the proposed method for blood count parameters

Parameter	Maximum difference between the two means	95% Lower confidence bound of difference between means	95% Upper confidence bound of difference between means	Stable
Haematocrit (HCT)	0.01	-0.01	0.00	no
Haemoglobin (HB)	1.53	-0.40	1.65	no
Red Blood Cells	0.05	-0.03	0.03	yes
Reticulocytes (%RBC)	0.09	-0.08	0.05	yes
Reticulocytes	4.58	-4.11	2.12	yes
Thrombocytes	6.17	-3.47	4.27	yes
White Blood Cells	0.09	-0.07	0.05	yes
Mean Corpuscular Volume (MCV)	1.49	-2.95	-1.02	no

Fig 3. Thrombocytes: the confidence interval of difference between means is within the maximum difference limit minimising the risk of false flagging. The difference in means between the two days can be said insignificant. The stability of this parameter can be accepted.

MCV: the confidence interval of difference between means is largely outside the maximum difference limit minimising the risk of false flagging. The difference in means between the two days cannot be said insignificant. In this case, the impact of sample instability on false flagging can not be eliminated.

HB: the confidence interval of difference between means is almost, but not quite, within the maximum difference limit. This case is borderline.



- Lack of stability was observed for haematocrit, MCV, and to a lesser extent haemoglobin (fig 4). For all other parameters, proof of sufficient stability was found.
- A meta-analysis has recently confirmed that 8h after collection, haematocrit and MCV begin to change significantly in samples at room temperature, while other haemogram parameters exhibit up to 3 days no difference over time [1].

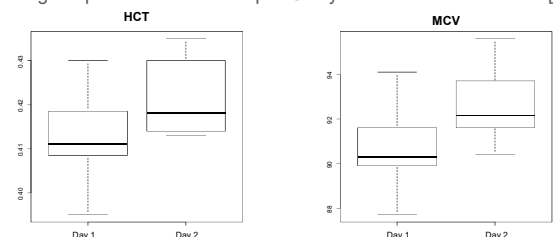


Fig 4. Box-plots of the haematocrit and MCV means for day 1 and day 2

REFERENCES

1. Wu DW, Li Y, Wang F. How Long can we Store Blood Samples: A Systematic Review and Meta-Analysis. *EBioMedicine*. 2017;24:277–85